



BHARATIYA ANTARIKSH HACKATHON 2025

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Team Name : CodeMind

Team Leader Name : Iqra Sayed

Problem Statement : Monitoring Air Pollution from Space using Satellite Observations, Ground-Based Measurements, Reanalysis Data, and AI/ML Techniques

Team Members

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Team Member-2: Name: Vivek Kumar Maurya College: A.P. Shah Institute of Technology	Team Member-3: Name: Pranav Ghodke College: A.P. Shah Institute of Technology

Brief about the Idea:

AI-Driven Air Quality Monitoring System

Using Satellite Data + AI for PM_{2.5}/PM₁₀ Prediction

Objective:

Predict city-level particulate matter (PM_{2.5}/PM₁₀) concentrations by fusing:

- **Satellite AOD** (INSAT-3D)
- **Ground sensors** (CPCB)
- **Weather data** (MERRA-2)

How It Works:

1. **Data Fusion:** Integrates AOD, weather (wind, PBLH), and historical PM.
2. **AI Model:** XGBOOST predicts PM levels with **<5 $\mu\text{g}/\text{m}^3$ MAE** (validated against ground data).
3. **Visualization:** Interactive maps and dashboards show real-time PM/AQI.

Opportunity should be able to explain the following:

How different is it from any of the other existing ideas?

How will it be able to solve the problem?

Current Approaches	Our AI Solution
✗ Relies on isolated ground sensors	✓ Combines satellite AOD + ground data
✗ Manual AOD-PM correlation	✓ Automated ML conversion (XGBoost)
✗ Static reports	✓ Time-aware predictions
✗ One national model	✓ City-specific tuned models

"Bridges the satellite-ground gap with AI that learns local AOD-PM relationships, weighted by real-time weather and time patterns."

1. Fills Spatial Gaps

- Uses **satellite AOD (TOTEXTTAU)** to estimate PM where ground sensors are missing
- *Code Proof:* AOD_PBLH_INTERACTION feature in XGBOOST

2. Adapts to Local Conditions

- Auto-adjusts for weather (WIND_SPEED, PBLH) and time-of-day (HOUR)
- *Code Proof:* METEOROLOGY_SCORE and temporal features

3. City-Specific Accuracy

- Separate models for Delhi/Mumbai/Ramanathapuram
- *Code Proof:* City-filtered training data

4. Faster Than Manual Methods

- Automated pipeline from raw data to predictions then to dashboard
- *Code Proof:* Batch processing in load_and_preprocess()

USP of the proposed solution

-  **Space-Based** - Uses INSAT-3D satellite AOD data
-  **AI-Calibrated** - Self-improving XGBOOST models
-  **Weather-Smart** - Considers wind/PBLH
-  **Time-Aware** - Hourly trend detection
-  **Hyperlocal** - City-specific model tuning
-  **Gap-Free** - Covers ground sensor blind spots

List of features offered by the solution:

Key Features of the Solution

Core Capabilities

- **Satellite-Powered PM Estimates**

Converts INSAT-3D AOD data into PM_{2.5}/PM₁₀ readings

- **City-Specific AI Models**

Customized predictions for Delhi, Mumbai, Ramanathapuram

- **Weather-Adjusted Accuracy**

Accounts for wind, temperature, and atmospheric conditions

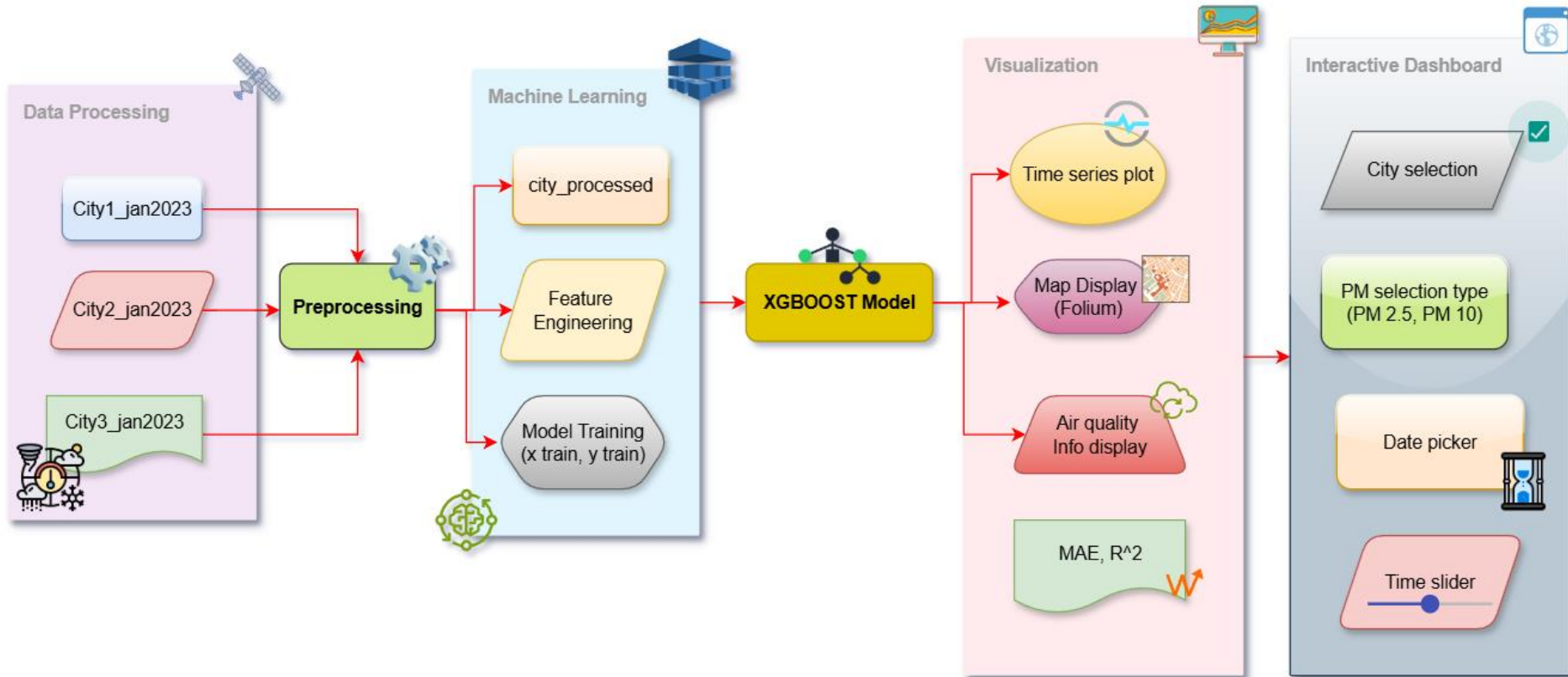
- **Time-Aware Predictions**

Detects hourly/daily pollution patterns

- **Near-Real-Time Updates**

Limited by satellite pass frequency

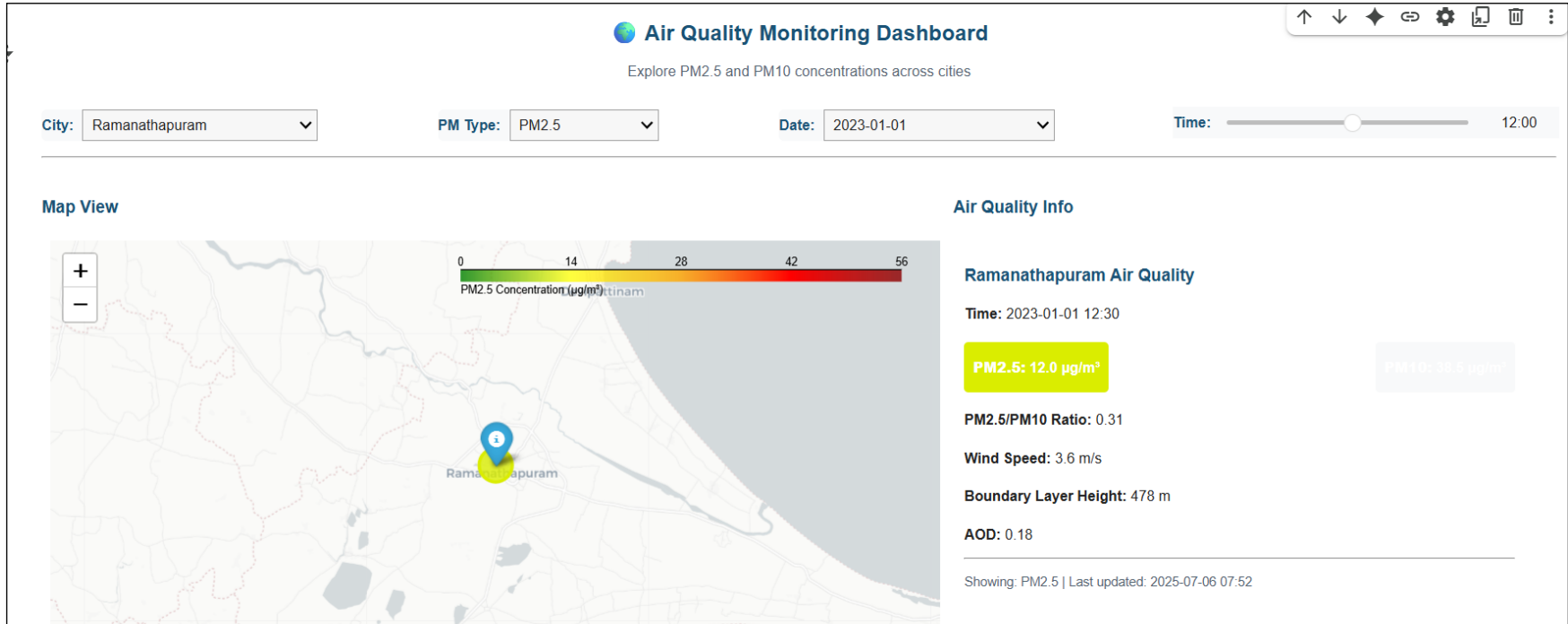
Process flow diagram or Use-case diagram:



Technologies to be used in the solution:

Technology Stack		
Component	Technology	Application
Satellite Data	INSAT-3D/3DR AOD	Aerosol measurements
Ground Truth	CPCB stations	PM validation
Weather Data	MERRA-2	Meteorology inputs
AI/ML Framework	XGBoost, Scikit-learn	Prediction models
Visualization	Folium, Matplotlib	Maps & plots

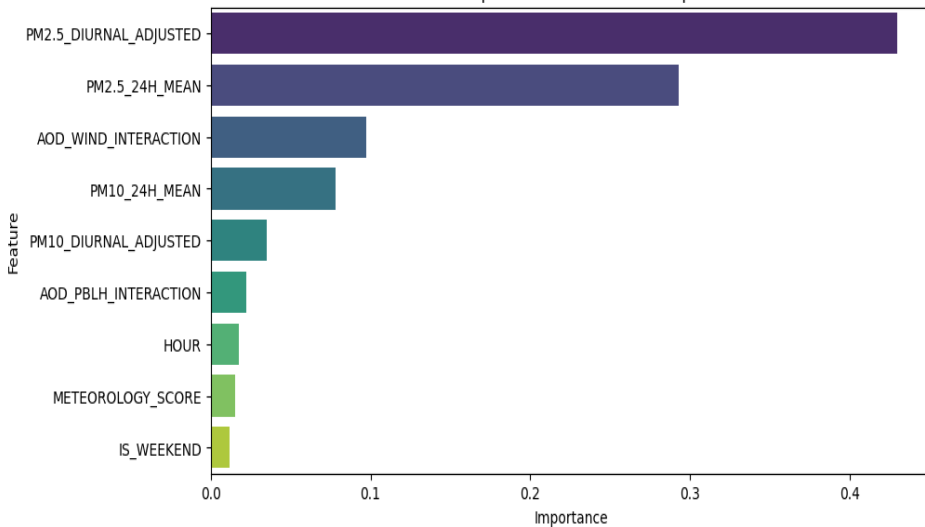
Wireframes/Mock diagrams of the proposed solution:



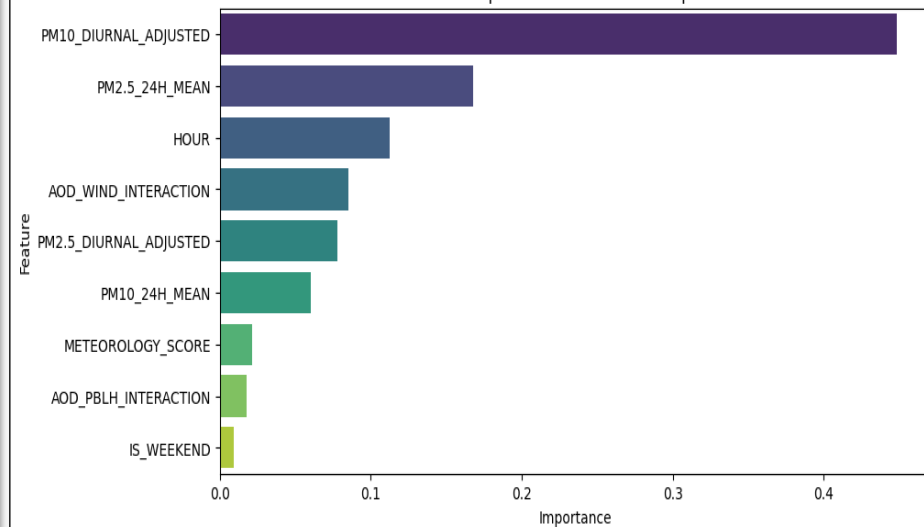
Predict PM2.5/PM10 levels for a specific city using satellite AOD data, ground measurements, and AI/ML, along with timestamps and air quality index (AQI) classification.

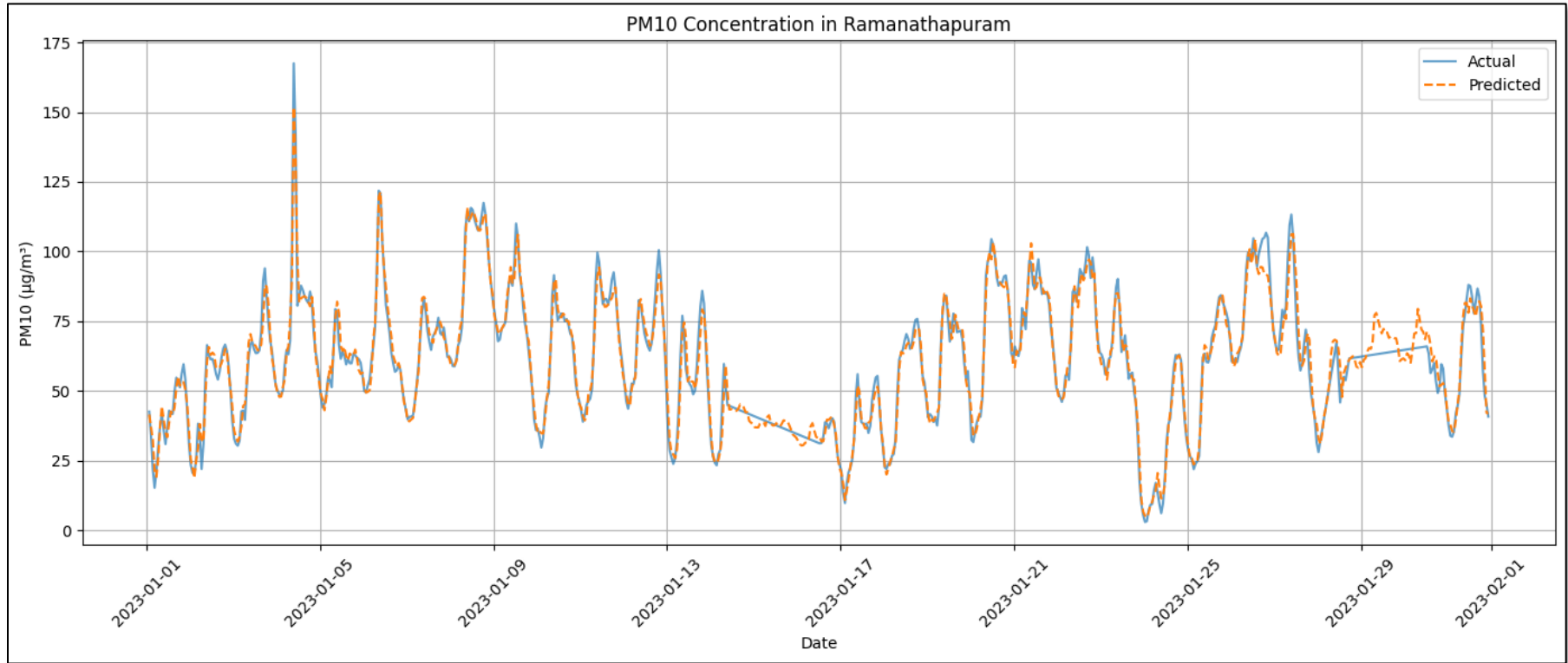
PM IMPORTANCE GRAPH FOR PM 2.5 & PM 10

Ramanathapuram - PM2.5 Feature Importance

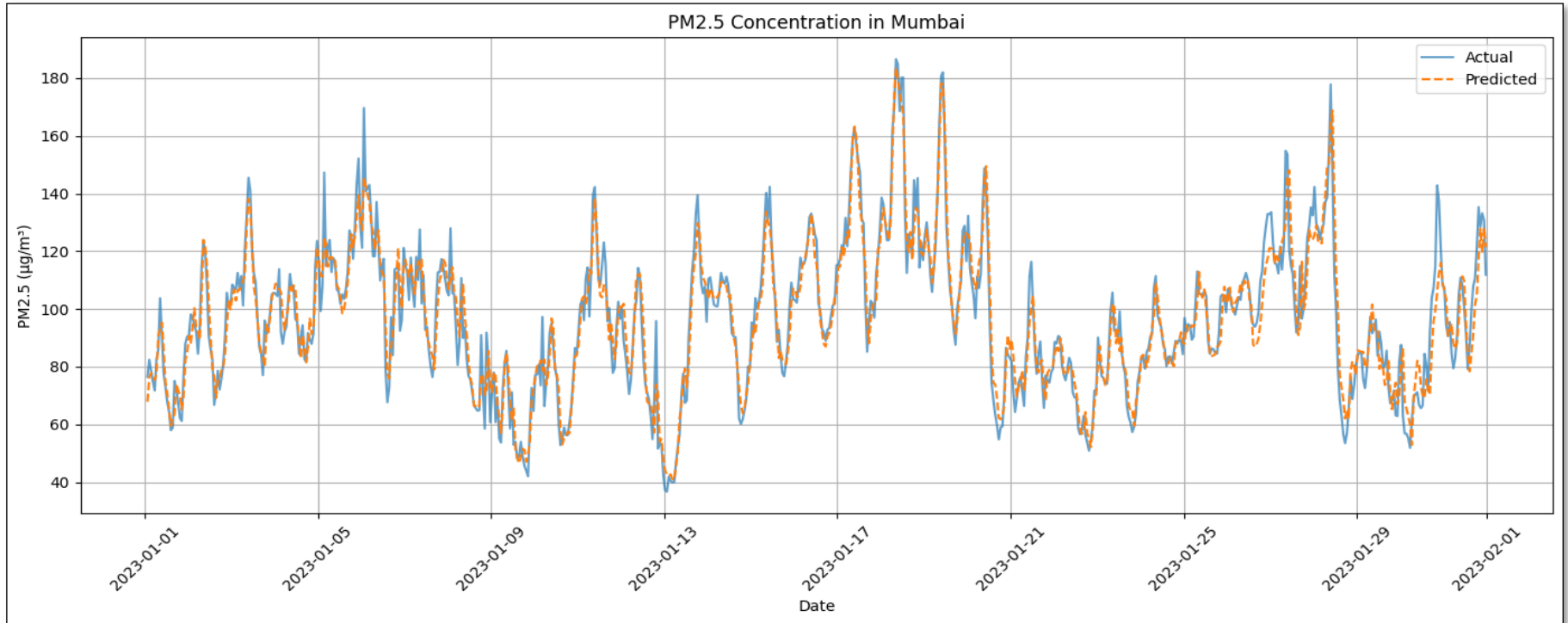


Ramanathapuram - PM10 Feature Importance

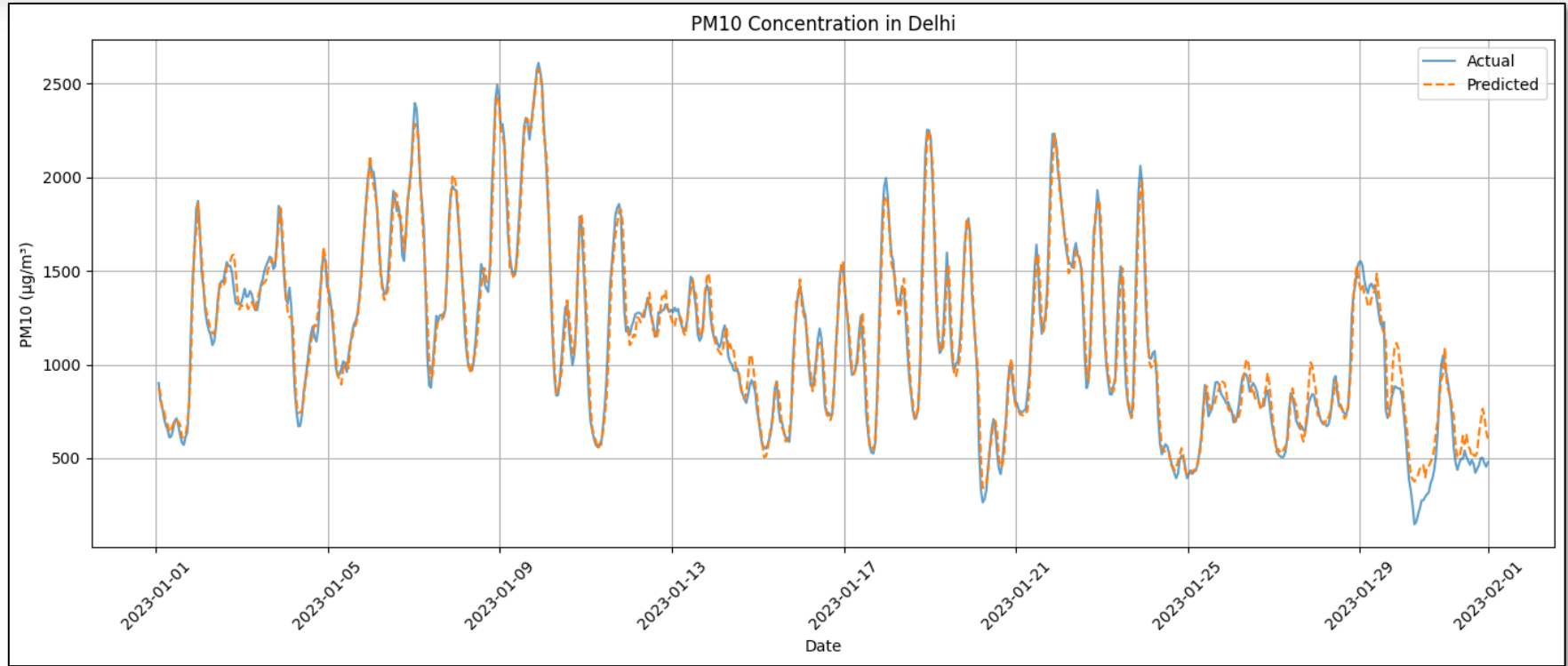




PM concentration map showing the ground values and the predicted values



PM 2.5 concentration map of Mumbai



PM 10 concentration map of Delhi

Demo Video Link-

<https://drive.google.com/file/d/13N-wn8tD6OeHzxF26hLTmu6HVTLUQ8Da/view?usp=drivesdk>

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THANK YOU

